

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and executed **DFT (Design for Testability)** strategies using **Python** and **Java** automation scripts to reduce test cost by **18%** and improve silicon yield analysis across multi-site **ATE (Automated Test Equipment)** platforms for enterprise mixed-signal product portfolios.
- Designed innovative **Probe and Test solutions** on high-performance **mixed-signal ATE platforms** using **Visual Basic** and **C** for hardware characterization, silicon functionality validation, and production-ready test methodology optimization, supporting ultra high-volume product ramps.
- Executed pre-tapeout **verification of Test solutions** using **Design Simulation tools** and **Python**-based test harnesses to validate digital and analog circuit behavior, reducing silicon debug cycles by **22%** and accelerating time-to-market for next-generation sensing products.
- Performed **Silicon functionality and debug, Characterization, and Yield Analysis** on production silicon using **Java Spring Boot** test frameworks and **SQL** (PostgreSQL, Oracle) data warehousing to identify failure modes, optimize measurement methodologies, and release offshore production solutions.
- Led **test development phase** execution including solution optimization, failure mode analysis, and continuous advancement of **measurement methodologies** through cross-functional collaboration with analog electronics design teams, delivering **3 major product test releases** on schedule.
- Applied strong **analytical skills** in signal processing and mixed-signal characterization using **Python** data analysis, **statistical methods**, and **R** regression analysis to drive improvement initiatives; demonstrated self-motivation and initiative in identifying and solving challenging technical problems in high-performance semiconductor test design.

SKILLS

Test Development & DFT – DFT definition and development, Design for Testability, Probe design, multi-site Test solutions, ATE platforms, silicon characterization, yield analysis.

Analog & Digital Electronics – Mixed-signal electronics, analog signal processing, digital circuit design, measurement methodologies, silicon debug, failure analysis.

Test Simulation & Verification – Design Simulation tools, pre-tapeout verification, Python test automation, Java Spring Boot test frameworks, C test harnesses, Visual Basic.

Programming & Scripting – Python, Java, C, Visual Basic, SQL (PostgreSQL, Oracle), shell scripting, CI/CD (Jenkins), Git.

Data Analysis & Characterization – Statistical analysis, R / RStudio, regression analysis, exploratory data analysis, time-series analysis, measurement data modelling, yield optimization.

Hardware & Production Support – Silicon functionality validation, hardware debug, production release coordination, change management, cross-functional team collaboration.

Cloud & Monitoring Infrastructure – AWS (EC2, CloudWatch), Docker containerization, Splunk monitoring, operational alerting and SLA management.

EDUCATION AND TRAINING

PROJECTS

Mixed-Signal Test Characterization & DFT Optimization Platform (HCL Technologies)

- Architected and implemented a **comprehensive DFT definition and development framework** using **Python** and **Java Spring Boot** to automate probe design validation, test pattern generation, and silicon characterization workflows for next-generation capacitive sensing and optical image stabilization IC designs.
- Developed **Design Simulation tools integration** and pre-tapeout verification harnesses in **C** and **Visual Basic** to validate test solution performance on high-performance **mixed-signal ATE platforms**, reducing simulation-to-silicon discrepancies by **28%** and accelerating design-to-production timeline by **6 weeks**.
- Built **Yield Analysis and measurement methodology optimization** dashboards using **Python** (pandas, NumPy, scikit-learn) and **SQL** data warehousing (PostgreSQL, Oracle) to identify failure modes, drive continuous improvement initiatives, and support ultra high-volume product ramp with **99.2% production test yield**.
- Led silicon **functionality and debug** execution across **3 major ATE multi-site test releases**, coordinating hardware characterization, analog and digital validation, and offshore production release support; demonstrated strong analytical skills and initiative in solving complex mixed-signal measurement challenges.
- Established cross-functional collaboration protocols with analog electronics design teams and test engineering leadership to drive innovation in **measurement methodologies**, deliver **4 improvement initiatives** per quarter, and mentor junior test engineers in DFT best practices and production test optimization.

Enterprise Production Monitoring & Operational Reliability Dashboard (HCL Technologies)

- Engineered a production-grade **monitoring and alerting solution** leveraging **Splunk**, **CloudWatch**, and **Python** automation to track system health, SLA compliance, and incident response times across distributed **AWS** infrastructure (EC2, S3, Lambda), reducing mean time to resolution (MTTR) by **35%**.
- Integrated **Design Simulation tool outputs** with operational dashboards to correlate test characterization data, silicon metrics, and production yield signals using **SQL** (PostgreSQL, Oracle) data pipelines and **statistical analysis** in **R**, enabling real-time visibility into product quality metrics.
- Automated **measurement methodology tracking** and continuous improvement workflows using **Jenkins CI/CD pipelines**, **Git** version control, and **Python** scheduled reporting to consolidate cross-site test data and drive organizational advancement of test practices across **12 product families**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)